

Griffiths Introduction to Quantum Mechanics 3rd edition Problem 4.21

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Take L_x as an example, by Equation (3.16), we need to show

$$\langle f|L_x f\rangle = \langle L_x f|f\rangle \quad (1)$$

. According to Equation (3.19), we have

$$\langle fx|p_z f\rangle = \langle p_z x f|f\rangle = \langle xp_z f|f\rangle$$

and

$$\langle fz|p_y f\rangle = \langle p_y z f|f\rangle = \langle zp_y f|f\rangle$$

. The left side of equation 1 = $f^*(xp_z - zp_y)f = \langle fx|p_z f\rangle - \langle fz|p_y f\rangle = \langle xp_z f|f\rangle - \langle zp_y f|f\rangle = f(xp_z f)^* - f(zp_y f)^* = f[(xp_z - zp_y)f]^* = [(xp_z - zp_y)f]^* f = (L_x f)^* f =$ Right side of equation 1.

It proved L_x , L_y and L_z is Hermitian according to Equation (3.16).

By Equation (3.20), we need to show

$$\langle f|L_{\pm} g\rangle = \langle L_{\mp} f|g\rangle \quad (2)$$

. Since L_x and L_y are Hermitian, so by Equation (3.17), we have

$$\langle f|L_x g\rangle = \langle L_x f|g\rangle$$

and

$$\langle f|L_y g\rangle = \langle L_y f|g\rangle$$

. By Equation (4.105),

The Left side of equation 2 = $\langle f|(L_x \pm iL_y)g\rangle = \langle f|L_x g\rangle \pm i\langle f|L_y g\rangle = \langle L_x f|g\rangle \pm i\langle L_y f|g\rangle = (L_x f)^* g \pm i(L_y f)^* g = (L_x f \mp iL_y f)^* g = (L_x \mp iL_y)^* g = \langle L_x \mp iL_y|g\rangle =$ the right side of equation 2.

So L_+ and L_- are each other's Hermitian conjugate. By Equation (3.20) we have

$$\langle f_l^m|L_+ f_l^{m-1}\rangle = \langle L_- f_l^m|f_l^{m-1}\rangle \quad (3)$$

$$\begin{aligned}
& \text{The left side of equation 3} = \iiint_{\text{all space}} f_l^{m*} L_+ f_l^{m-1} d\mathcal{V} = \iiint_{\text{all space}} f_l^{m*} (A_l^{m-1}) f_l^{m-1} d\mathcal{V} \\
& = A_l^{m-1} \iiint_{\text{all space}} f_l^{m*} f_l^m d\mathcal{V} = A_l^{m-1} \iiint_{\text{all space}} |f_l^m|^2 d\mathcal{V} = A_l^{m-1}
\end{aligned}$$

The last step is because eigenfunctions like f_l^m are normalized.

$$\begin{aligned}
& \text{Similarly, the right side of equation 3} = \iiint_{\text{all space}} (L_- f_l^m)^* f_l^{m-1} d\mathcal{V} = \iiint_{\text{all space}} [(B_l^m) f_l^{m-1}]^* f_l^{m-1} d\mathcal{V} \\
& = \iiint_{\text{all space}} B_l^{m*} f_l^{m-1*} f_l^{m-1} d\mathcal{V} = B_l^{m*} \iiint_{\text{all space}} |f_l^{m-1}|^2 d\mathcal{V} = B_l^{m*} \\
& , \text{ which shows}
\end{aligned}$$

$$A_l^{m-1} = B_l^{m*} \quad (4)$$

On the other hand, since L_+ and L_- are each other's Hermitian conjugate, so $L_+ L_-$ is Hermitian by Problem 3.5 because $(L_+ L_-)^* = L_+ L_-$. By Equation (4.112) and Equation (4.118),

$$L_+ L_- f_l^m = (L^2 - L_z^2 + \hbar^2 L_z) f_l^m = L^2 f_l^m - L_z^2 f_l^m + \hbar^2 L_z f_l^m = \hbar^2 l(l+1) f_l^m - \hbar^2 m^2 f_l^m + \hbar^2 m f_l^m = B_l^m A_l^{m-1} f_l^m$$

. So,

$$\hbar^2 l(l+1) - \hbar^2 m^2 + \hbar^2 m = B_l^m A_l^{m-1} \quad (5)$$

. By equation 4 and equation 5, we get

$$|A_l^m| = \hbar \sqrt{l(l+1) - m(m+1)} = \hbar \sqrt{(l-m)(l+m+1)}$$

and

$$|B_l^m| = \hbar \sqrt{l(l+1) - m(m-1)} = \hbar \sqrt{(l+m)(l-m+1)}$$

. Since the phase of A_l^m and B_l^m is insignificant, and by equation 4, we can set them to be real.

$$A_l^m = \hbar \sqrt{l(l+1) - m(m+1)} = \hbar \sqrt{(l-m)(l+m+1)}$$

and

$$B_l^m = \hbar \sqrt{l(l+1) - m(m-1)} = \hbar \sqrt{(l+m)(l-m+1)}$$

. By these results, we can find that f_l^l is the eigenfunction of L_+ with eigenvalue 0 and f_l^{-l} is the eigenfunction of L_- with eigenvalue 0.